2 Real Analysis 2023
Tutorial 6

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Series

Problem 21 Prove or disprove:

(a) If $\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty}(a_n + b_n)$ diverges.

(b) Assume that $a_n > 0$ for all $n \in \mathbb{N}$. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} \frac{1}{a_n}$ diverges.

(c) Assume that $a_n > 0$ for all $n \in \mathbb{N}$. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} \frac{1}{a_n}$ converges.

Solution.

(a) True. Assume, by contradiction, that $\sum_{n=1}^{\infty}(a_n + b_n)$ converges. Then

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty}(a_n + b_n) - \sum_{n=1}^{\infty} a_n$$

is a difference of two convergent series. Consequently, $\sum_{n=1}^{\infty} b_n$ converges, which contradicts the assumption. This proves that $\sum_{n=1}^{\infty}(a_n + b_n)$ diverges.

(b) True. Since $\sum_{n=1}^{\infty} a_n$ converges, we have $\lim_{n\to\infty} a_n = 0$. But then the sequence $\left(\frac{1}{a_n}\right)_{n=1}^{\infty}$ is unbounded and, in particular, does not tend to zero. It follows that $\sum_{n=1}^{\infty} \frac{1}{a_n}$ diverges.

(c) False. Both series $\sum_{n=1}^{\infty} n$ and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverge.

Problem 22 For each of the following series, determine whether it converges or diverges.

(a) $\displaystyle\sum_{n=3}^{\infty} \frac{n+4}{n^2 - 3n + 1}$, (b) $\displaystyle\sum_{n=0}^{\infty} \frac{\sqrt{n+1}}{n^2+1}$, (c) $\displaystyle\sum_{n=1}^{\infty} \frac{3+\sin n}{n^3}$,

(d) $\displaystyle\sum_{n=0}^{\infty} \sqrt{\frac{n+1}{n^2+1}}$, (e) $\displaystyle\sum_{n=1}^{\infty} \frac{1+2n}{3n}$, (f) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(2n-5)}$,

(g) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n2^n}$, (h) $\displaystyle\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^n$, (i) $\displaystyle\sum_{n=2}^{\infty} \frac{\ln n}{n^3}$.

Solution.

(a) For all $n \geq 3$
$$\frac{n+4}{n^2-3n+1} \geq \frac{n}{n^2+1} \geq \frac{n}{n^2+n^2} = \frac{1}{2n}.$$
The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, hence $\sum_{n=1}^{\infty} \frac{1}{2n}$ diverges. By the Comparison Test, $\sum_{n=3}^{\infty} \frac{n+4}{n^2-3n+1}$ diverges.

(b) For all $n \geq 1$
$$\frac{\sqrt{n+1}}{n^2+1} \leq \frac{\sqrt{n+n}}{n^2} = \frac{\sqrt{2}\sqrt{n}}{n^2} = \frac{\sqrt{2}}{n^{\frac{3}{2}}}.$$
The series $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ converges (this is a p-series with $p = \frac{3}{2} > 1$), hence $\sum_{n=1}^{\infty} \frac{\sqrt{2}}{n^{\frac{3}{2}}}$ converges. By the Comparison Test, $\sum_{n=0}^{\infty} \frac{\sqrt{n+1}}{n^2+1}$ converges.

(c) Notice that $\frac{3+\sin n}{n^3} \geq \frac{3-1}{n^3} = \frac{2}{n^3} > 0$, so that we may apply the Comparison Test. Now for all $n \geq 1$ we have
$$\frac{3+\sin n}{n^3} \leq \frac{3+1}{n^3} = \frac{4}{n^3},$$
and the series $\sum_{n=1}^{\infty} \frac{4}{n^3}$ converges (this is the convergent p-series with $p = 3 > 1$, multiplied by 4). By the Comparison Test, $\sum_{n=1}^{\infty} \frac{3+\sin n}{n^3}$ converges.

(d) For $n \geq 1$
$$\sqrt{\frac{n+1}{n^2+1}} \geq \sqrt{\frac{n}{n^2+n^2}} \geq \sqrt{\frac{1}{2n}} = \frac{1}{\sqrt{2}\sqrt{n}}.$$
The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2}\sqrt{n}}$ (this is a divergent p-series with $p = \frac{1}{2} \leq 1$, multiplied by $\frac{1}{\sqrt{2}}$). By the Comparison Test, $\sum_{n=0}^{\infty} \sqrt{\frac{n+1}{n^2+1}}$ diverges.

(e) We have
$$\lim_{n\to\infty} \frac{1+2n}{3n} = \lim_{n\to\infty} \frac{\frac{1}{n}+2}{3} = \frac{2}{3} \neq 0.$$
Therefore, the series $\sum_{n=1}^{\infty} \frac{1+2n}{3n}$ diverges.

(f) For $n \geq 3$
$$0 \leq \frac{1}{n(2n-5)} \leq \frac{1}{n(2n-n)} = \frac{1}{n^2}.$$
The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. By the Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{n(2n-5)}$ converges.

(g) For $n \geq 1$
$$\frac{1}{n2^n} \leq \frac{1}{2^n} = \left(\frac{1}{2}\right)^n.$$
The series $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ is a convergent geometric series. By the Comparison Test, $\sum_{n=1}^{\infty} \frac{1}{n2^n}$ converges.

(h) We have

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e \neq 0.$$

Therefore, the series $\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^n$ diverges.

(i) For $n \geq 2$

$$\frac{\ln n}{n^3} \leq \frac{n}{n^3} = \frac{1}{n^2}.$$

The series $\sum_{n=1}^{\infty}\frac{1}{n^2}$ converges. By the Comparison Test, $\sum_{n=2}^{\infty}\frac{\ln n}{n^3}$ converges.

Subsequences

Problem 23

(a) Describe all convergent subsequences of the sequence

$$1, -1, 1, -1, 1, -1, \ldots.$$

Hint: There are infinitely many convergent subsequences, but they can have only two possible limits.

(b) Give an example of a sequence that has convergent subsequences with three different limits.

Solution.

(a) The convergent subsequences are exactly all possible eventually constant subsequences. I.e., they start with terms 1 and -1 in an arbitrary order, but end with a constant sequence of either 1's or -1's.

(b) For example,

$$0, 1, 2, 0, 1, 2, 0, 1, 2, \ldots.$$

Problem 24 Prove the following statement:

Let $(a_n)_{n=1}^{\infty}$ be a sequence. If $\lim_{k\to\infty} a_{2k} = \lim_{k\to\infty} a_{2k-1} = L$, then $\lim_{n\to\infty} a_n = L$.

Solution.

Given $\varepsilon > 0$, there exists $N_0, N_1 \in \mathbb{N}$ such that

$$|a_{2k} - L| < \varepsilon \quad \text{for all } 2k \geq N_0,$$

and

$$|a_{2k-1} - L| < \varepsilon \quad \text{for all } 2k-1 \geq N_1.$$

Put $N = \max\{N_0, N_1\}$, then for all $n \geq N$

$$|a_n - L| < \varepsilon.$$

This proves that $\lim_{n\to\infty} a_n = L$.